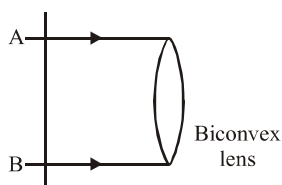


WAVE OPTICS

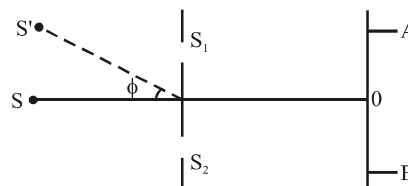
1. Wave theory of light is not initially accepted because :-
 - (1) It does not explain reflection and refraction processes
 - (2) It does not explain photoelectric effect
 - (3) It does not explain doppler's effect
 - (4) It does not explain propagation of light through vacuum
2. According to Maxwell light waves are associated with :-
 - (1) a constant magnitude electric field
 - (2) a magnetic field of constant magnitude
 - (3) changing electric and magnetic fields
 - (4) electric and magnetic fields of constant magnitude
3. A wavefront is a :-
 - (1) a surface imagined parallel and coplaner with light rays
 - (2) a surface around a source such that each point of it is at a constant distance from the source
 - (3) a surface which contains the plane of oscillations of electric field of light
 - (4) a surface which is created by medium particles oscillating in same phase
4. If AB is incident wavefront. Then refracted wavefront (just after transmission) is :-



- (1) (2) (3) (4)

5. Two source are said to be coherent when :-
 - (1) Phase difference between waves produced by them is a function of time
 - (2) Phase difference between waves produced by them is $\pi/2$
 - (3) Phase difference between waves produced by them is 0
 - (4) Phase difference between them remains constant with time

6. In the phenomenon of interference, energy is :-
 - (1) Destroyed at destructive interference
 - (2) Created at constructive interference
 - (3) Conserved but it is redistributed
 - (4) Same at all points
7. The shape of the fringe obtained on the screen in case of YDSE is :-
 - (1) a straight line
 - (2) a parabola
 - (3) a hyperbola
 - (4) a circle
8. In young's double slit experiment, if source S is shifted by an angle ϕ as shown. Then central-bright fringe will be shifted by angle ϕ towards:-



- (1) end A of screen
 - (2) end B of screen
 - (3) does not shift at all
 - (4) either end A or B depending on extra phase difference caused by shifting of source
9. In young's double-slit experiment the central bright fringe can be identified.
 - (1) as it has greater intensity than the other bright fringe
 - (2) as it is wider than the other bright fringes
 - (3) as it is narrower than the other bright fringe
 - (4) by using white light instead of monochromatic light
10. Diffraction is a general characteristic exhibited by:-
 - (1) Sound waves
 - (2) Light waves
 - (3) Water waves
 - (4) All of the above
11. For better resolution, a telescope must have a :-
 - (1) Large diameter objective
 - (2) Small diameter objective
 - (3) May be large
 - (4) Neither large nor small
12. The resolving power of a microscope is basically determined by the :-
 - (1) Speed of the light used
 - (2) Wavelength of the light used
 - (3) Both (1) and (2)
 - (4) Neither (1) nor (2)

- 13.** Which of the given statement is/are correct for phenomenon of diffraction ?
- For diffraction through a single-slit the wavelength of wave must be comparable to the size of the slit
 - The diffraction is very common in sound wave but not so common in light waves
 - Diffraction is only observed in electromagnetic waves
- Only i
 - ii and iii
 - i and ii
 - i, ii, and iii
- 14.** The phenomenon of polarisation of light indicates that :-
- light is a longitudinal wave
 - light is a transverse electromagnetic wave
 - light is a transverse wave only
 - either (2) or (3)
- 15.** Which factor could possibly influence the speed of light in vacuum :-
- Motion of the source or observer
 - Frequency of light/wave length of light
 - Intensity of light wave
 - None of the above
- 16.** Which factor could possibly influence the speed of light in isotropic medium :-
- Direction of propagation
 - Nature of source
 - Motion of source relative to the medium
 - Motion of the observer relative to the medium
- 17.** When scattering of light is taking place and observer is observing at 90° scattered light will be :-
- Unpolarized
 - Partially polarised
 - Circularly polarised
 - Perfectly plane polarised
- 18.** When plane polarized light having plane of incident as plane of vibration is incident at Brewster's angle :-
- reflected light will be polarised
 - There will be no reflection
 - there will be no refraction
 - none of the above
- 19.** When scattering sunlight is being caused by air molecule in sky it radiate energy :-
- Equally in all direction
 - Maximum energy in the direction of oscillations
 - No energy in the direction of oscillations
 - Does not radiate energy at all
- 20.** Beyond Fresnel distance :-
- Diffraction becomes insignificant
 - Diffraction are smaller compared to the size of beam
 - Spreading of light due to diffraction dominates over ray optics
 - Ray optics is valid
- 21.** Mark incorrect option :-
- A telescope produces images of far object near to eye
 - A telescope produces images of far object magnified
 - A microscope produces images of near object magnified
 - A microscope produces image of near object far from eye
- 22.** Presence of oil in oil immersed microscope is to:-
- to increase magnification of microscope
 - increase aperture of microscope
 - increase numerical aperture of microscope
 - to make numerical aperture less than one in microscope
- 23.** Interference and diffraction effect are :-
- Inconsistent with energy conservation
 - Consistent with energy conservation
 - Destruction in region of darkness
 - Creation of energy in the region of brightness
- 24.** How we can increase number of interference fringes inside the central maxima of diffraction pattern of double slit :-
- by increasing distance of the screen
 - by increasing size of slits
 - by decreasing separation between slits
 - by decreasing size of slits

25. What is the effect on the interference fringes in YDSE when source slit moved closer to the double slit plane :-
 (1) β increases
 (2) β decrease
 (3) Fringes become less sharp
 (4) fringes become more sharp
26. What is the effect on interference fringes in YDSE when width of the source slit increased.
 (1) β increases
 (2) β decreases
 (3) Fringes become less sharp
 (4) Fringes become more sharp
27. For two light source to be coherent :-
 (1) their amplitude must be same
 (2) phase of wave should not change
 (3) phase difference between waves must remain constant
 (4) wave must be transverse
28. The principle of superposition of waves applies whenever two or more sources of light :-
 (1) are coherent
 (2) are of same frequency
 (3) obtained from same source
 (4) illuminate the same point
29. The equation of a light wave is written as $y = A \sin(kx - \omega t)$. Here y represents :-
 (1) displacement of ether particles
 (2) Pressure in the medium
 (3) density of medium
 (4) Electric field
30. Light waves emitted from an ordinary source change in time of order of :-
 (1) 10^{-2} sec (2) 10^{-8} sec
 (3) 10^{-4} sec (4) 1 sec
31. A plane wavefront is incident on concave mirror then spherical wavefront will converge at :-
 (1) focus (2) centre of curvature
 (3) infinity (4) $\frac{f}{2}$ point
32. What happens when two separate light source illuminate a wall :-
 (1) Uniform illumination
 (2) Interference fringes
 (3) Dark appear
 (4) Four times intensity of each source
33. A plane wavefront is incident on a prism then emerging wavefront will be :-
 (1) Plane
 (2) Spherical
 (3) Cylindrical
 (4) None
34. Which statement is wrong about interference and diffraction :-
 (1) In interference distance between bright and dark is same but in diffraction width of central maxima is double of other maxima
 (2) Interference pattern is superposition of two waves originating from two narrow slits
 diffraction pattern is superposition of a continuous family of waves originating from each point on a single slit
 (3) For a single slit of width a the first null of the interference pattern occurs at an angle of $\frac{\lambda}{a}$. At the same angle of $\frac{\lambda}{a}$ we get a maxima for two narrow slits separated by a distance a
 (4) Intensity is same for all bright band in interference and diffraction
35. In double slit experiment if one slit is blocked then what we get on screen.
 (1) Uniform brightness on screen
 (2) Diffraction pattern on screen
 (3) Interference pattern remain same
 (4) Intensity becomes half of Initial intensity
36. If a medium is placed between object and objective lens of microscope then resolving power of microscope will :-
 (1) Increase
 (2) Decrease
 (3) No change
 (4) Nothing can be predicted
37. Colour appear in CD is due to :-
 (1) Interference
 (2) Diffraction
 (3) Polarisation
 (4) None

38. If the light from an ordinary source passes through a polaroid sheet then intensity of light becomes:-
 (1) half
 (2) remain same
 (3) one fourth
 (4) Three fourth
39. An ordinary light is pass from a polaroid sheet P_1 . Then after P_1 another polaroid sheet P_2 is placed then. Intensity of light after P_2 will be :-
 (1) remain same
 (2) half
 (3) Zero
 (4) Change from zero to half of intensity of unpolarised light when P_2 is rotate perpendicular to direction of propagation
40. Light appears to travel in straight line since :-
 (1) It is not absorbed by the atmosphere
 (2) It is reflected by the atmosphere
 (3) Its wavelength is very small
 (4) Its velocity is very large
41. In a single slit diffraction experiment, the width of the slit is made double the original width. How does this affect the intensity of the central diffraction band ?
 (1) becomes double
 (2) becomes half
 (3) becomes one-fourth
 (4) becomes 4 times
42. When a tiny circular obstacle is placed in the path of light from distant source. What is observed at the centre of the shadow of the obstacle.
 (1) Dark spot
 (2) Bright spot
 (3) Dark & bright fringe
 (4) None
43. On which factor speed of light in a medium depends :-
 (1) Nature of source
 (2) Direction of propagation
 (3) Wavelegnth
 (4) None of these
44. When a low flying aircraft passes overhead a slight shaking picture appear on our T.V. screen. This is due to :-
 (1) Wind effect on antenna
 (2) T.V. gets vibrated due to aircraft pressure
 (3) Interference of signal
 (4) Polarization of signal
45. What will be the shape of wavefront when light emerging out of a convex lens for a point source placed at its focus ?
 (1) Spherical (2) Planar
 (3) Cylindrical (4) None
46. What will be the shape of wavefront for the portion of the wavefront of light from distant source.
 (1) Spherical (2) Planar
 (3) Cylindrical (4) None
47. Huygen's principle of secondary wavelets may be used to :-
 (1) Find new position of a wavefront
 (2) Explain snell's law
 (3) Explain law of reflection
 (4) All of these
48. Doppler effect of light is the basis for the measurement of :-
 (1) Radial velocities of distant galaxies
 (2) Velocity of light
 (3) Velocity of observer
 (4) None
49. In the wave picture of light, intensity of light is determined by the square of amplitude of the wave. What determines the intensity of light in the photon picture of light at a given frequency?
 (1) No. of photon crossing an unit area per unit time
 (2) Area of cross section only
 (3) Density of medium
 (4) All of these
50. Upto what distance ray optics is valid in diffraction.
 (1) $\frac{a^2}{\lambda}$ (2) $\frac{a}{\lambda^2}$
 (3) $\frac{a}{\lambda}$ (4) $\frac{a^2}{\lambda^2}$
 Where a is size of aperture

51. Consider sunlight incident on a pinhole of width $10^3 \text{ \AA}$. The image of the pinhole seen on a screen shall be :-
- (1) A sharp white ring
 - (2) Different from a geometrical image
 - (3) A diffused central spot, white in colour
 - (4) Diffused coloured region around a sharp central white spot
52. Consider the diffraction pattern for a small pinhole. As the size of the hole is increased :-
- (1) The intensity increases
 - (2) The intensity decreases
 - (3) The intensity remains constant
 - (4) Intensity will be disappeared